Pre-6.4 Midpoint, Distance, and Slope

Name _____ Period ____ Date _____

Geometry (30-minute plan) · Student sheet

I can…

- find the middle of a segment.
- find the length of a segment.
- find slopes and use them to compare lines.

Vocabulary · fill in the missing word

Midpoint — $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$: _____ the x's and the y's

Distance — $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$: the _____ Theorem

Slope — $m = \frac{y_2 - y_1}{x_2 - x_1}$: _____ over run

Parallel / perpendicular — same slope / slopes _____ to −1

Part 1 · Three questions, three formulas

Question	Formula	What it means
Where's the middle?	M =	
How long is it?	d =	
How steep is it?	m =	

Which formulas start by subtracting? _____ Which one starts by adding? _____

Example

A(−2, 1) and B(4, 9). Find the middle, the length, and the slope.

Part 2 · We do

Problem 1

P(−5, 2) and Q(3, 8). Find the midpoint, distance, and slope.

Problem 2

T(3, −2) and U(3, 6). Find the midpoint, distance, and slope.

Problem 3

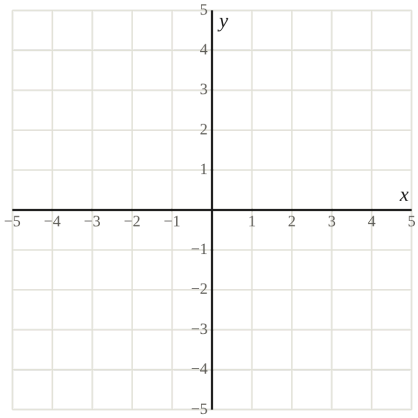
Segment 1 goes through (1, 2) and (4, 8). Segment 2 goes through (0, 5) and (4, 3). Are they parallel, perpendicular, or neither? Show the slopes.

Part 3 · Partner practice

Is ABCD a square?

A(−3, 1), B(1, 4), C(4, 0), D(0, −3). Plot ABCD.

Partner A: find the four side lengths. Partner B: find the four slopes. Then decide together.



Side	Length	Slope
AB		
BC		
CD		
DA		

Is ABCD a square? How do you know? _____